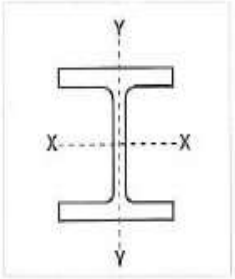


4b (ii) Columns (Metric Units) - Part I

Designation		Height Of Column	Width Of Column	Thickness		Root Radius	Depth Between Fillets	Area Of Section	Ratios For Local Buckling		Dimensions For Detailing		
Size	Unit Weight			Flange	Web				Flange	Web	End Clearance	Notch	
	M	H	B	T	t	r	d	A	b/T	d/t	C	N	n
mm	kg/m	mm	mm	mm	mm	mm	mm	cm^2			mm	mm	mm
100 x 100	14.8	100	100	7	5	10	66	19.2	7.14	13.2	5	58	17
	16.9	100	100	8	6	8	68	21.6	6.25	11.3	5	57	16
	17.2	100	100	8	6	10	64	21.9	6.25	10.7	5	57	18
125 x 125	23.6	125	125	9	6.5	8	91	30.0	6.94	14.0	5	69	17
	23.8	125	125	9	6.5	10	87	30.3	6.94	13.4	5	69	19
150 x 150	31.1	150	150	10	7	8	114	39.6	7.50	16.3	6	82	18
	31.5	150	150	10	7	11	108	40.1	7.50	15.4	6	82	21
	37.4	154	151	12	8	11	108	47.7	6.29	13.5	6	82	23
175 x 175	32.8	171	174	9	6	12	129	41.7	9.67	21.5	5	94	21
	40.2	175	175	11	7.5	12	129	51.2	7.95	17.2	6	94	23
	40.4	175	175	11	7.5	13	127	51.4	7.95	16.9	6	94	24
200 x 200	41.4	196	199	10	6.5	13	150	52.7	9.95	23.1	5	106	23
	49.9	200	200	12	8	13	150	63.5	8.33	18.8	6	106	25
	57.8	204	201	14	9	13	150	73.6	7.18	16.7	7	106	27
	65.7	208	202	16	10	13	150	83.7	6.31	15.0	7	106	29
250 x 250	66.5	248	249	13	8	16	190	84.7	9.58	23.8	6	131	29
	71.8	250	250	14	9	13	196	91.4	8.93	21.8	7	131	27
	72.4	250	250	14	9	16	190	92.2	8.93	21.1	7	131	30
	98.1	260	253	19	12	16	190	125	6.66	15.8	8	131	35
300 x 300	87.0	298	299	14	9	18	234	111	10.7	26.0	7	155	32
	93.0	300	300	15	10	13	244	118	10.0	24.4	7	155	28
	94.0	300	300	15	10	18	234	120	10.0	23.4	7	155	33
	106	304	301	17	11	18	234	135	8.85	21.3	8	155	35
	125	310	303	20	13	18	234	159	7.58	18.0	9	155	38
	130	310	305	20	15	18	234	165	7.63	15.6	10	155	38
	147	312	310	21	20	18	234	187	7.38	11.7	12	155	39
350 x 350	113	344	348	16	10	13	286	144	10.9	28.6	7	179	29
	115	344	354	16	10	20	272	148	11.1	27.2	7	182	36
	135	350	350	19	12	13	286	172	9.21	23.8	8	179	32
	137	350	350	19	12	20	272	174	9.21	22.7	8	179	39
	159	356	352	22	14	20	272	202	8.00	19.4	9	179	42
	181	362	354	25	16	20	272	230	7.08	17.0	10	179	45
400 x 400	140	388	402	15	15	22	314	178	13.4	20.9	10	204	37
	147	394	398	18	11	22	314	187	11.1	28.5	8	204	40
	172	400	400	21	13	22	314	219	9.52	24.2	9	204	43
	200	406	403	24	16	22	314	255	8.40	19.6	10	204	46
	232	414	405	28	18	22	314	295	7.23	17.4	11	204	50
	283	428	407	35	20	22	314	361	5.81	15.7	12	204	57
	415	458	417	50	30	22	314	529	4.17	10.5	17	204	72
	605	498	432	70	45	22	314	770	3.09	7.0	25	204	92

Dimensions and Properties



4b (ii) Columns (Metric Units) - Part II

Surface Area Per Metre	Second Moment Of Area		Radius Of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant	Designation	
	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	x-x	y-y	u	x	H	J	Unit Weight	Size
m^2	cm^4	cm^4	cm	cm	cm^3	cm^3	cm^3	cm^3			cm^6	cm^4	kg/m	mm
0.570	344	117	4.20	2.47	69.0	23.0	78.0	36.0	0.843	12.3	0.00253	3.53	14.8	**100 x 100**
0.574	378	134	4.18	2.49	75.6	26.7	86.4	41.0	0.836	11.1	0.00283	4.73	16.9	
0.571	383	134	4.18	2.47	76.5	26.8	87.6	41.2	0.837	10.7	0.00283	5.17	17.2	
0.723	840	293	5.29	3.13	134	46.9	152	71.7	0.839	12.8	0.00987	7.91	23.6	**125 x 125**
0.720	847	294	5.29	3.11	136	47.0	154	71.9	0.840	12.5	0.00988	8.43	23.8	
0.872	1623	563	6.40	3.77	216	75.1	243	114	0.841	14.1	0.0276	12.5	31.1	**150 x 150**
0.867	1641	563	6.39	3.75	219	75.1	246	115	0.842	13.7	0.0276	13.5	31.5	
0.877	2018	690	6.51	3.80	262	91.4	298	140	0.845	11.8	0.0348	22.2	37.4	
1.01	2304	791	7.43	4.35	269	90.9	298	138	0.845	17.5	0.0519	11.5	32.8	**175 x 175**
1.01	2884	984	7.50	4.38	330	112	369	171	0.844	14.7	0.0662	20.4	40.2	
1.01	2895	984	7.50	4.37	331	112	370	172	0.845	14.5	0.0662	21.0	40.4	
1.15	3846	1315	8.54	5.00	392	132	433	201	0.847	18.2	0.114	17.6	41.4	**200 x 200**
1.16	4716	1602	8.62	5.02	472	160	525	244	0.846	15.5	0.142	29.8	49.9	
1.17	5603	1897	8.73	5.08	549	189	617	287	0.847	13.6	0.171	45.8	57.8	
1.18	6531	2201	8.83	5.13	628	218	710	332	0.848	12.2	0.203	66.7	65.7	
1.45	9931	3348	10.8	6.29	801	269	883	408	0.850	17.9	0.462	46.7	66.5	**250 x 250**
1.46	10750	3648	10.8	6.32	860	292	953	443	0.847	17.1	0.508	55.8	71.8	
1.45	10830	3649	10.8	6.29	867	292	960	444	0.848	16.7	0.508	58.7	72.4	
1.48	15340	5134	11.1	6.41	1180	406	1330	618	0.849	12.8	0.745	141	98.1	
1.74	18850	6242	13.0	7.51	1265	418	1389	634	0.850	20.0	1.26	71.3	87.0	**300 x 300**
1.76	20190	6753	13.1	7.55	1346	450	1484	683	0.847	19.4	1.37	82.1	93.0	
1.75	20410	6756	13.1	7.51	1361	450	1501	684	0.848	18.8	1.37	88.1	94.0	
1.76	23380	7733	13.2	7.57	1538	514	1705	781	0.849	16.9	1.59	125	106	
1.78	28130	9282	13.3	7.64	1815	613	2031	932	0.849	14.6	1.95	200	125	
1.78	28630	9470	13.2	7.57	1847	621	2079	949	0.842	14.4	1.99	215	130	
1.79	31370	10450	13.0	7.48	2011	674	2295	1040	0.830	13.0	2.21	298	147	
2.04	32850	11240	15.1	8.84	1910	646	2092	978	0.848	21.1	3.02	111	113	**350 x 350**
2.05	33810	11840	15.1	8.95	1966	669	2153	1014	0.846	20.4	3.18	123	115	
2.05	39847	13583	15.2	8.89	2277	776	2515	1176	0.847	18.0	3.72	186	135	
2.04	40300	13590	15.2	8.84	2303	776	2545	1179	0.849	17.5	3.72	199	137	
2.06	47590	16000	15.3	8.90	2674	909	2979	1382	0.848	15.4	4.46	305	159	
2.07	55190	18500	15.5	8.96	3049	1045	3424	1591	0.848	13.7	5.25	445	181	
2.32	48970	16260	16.6	9.55	2524	809	2802	1237	0.830	22.5	5.66	156	140	**400 x 400**
2.32	56150	18930	17.3	10.1	2850	951	3118	1441	0.850	20.9	6.69	194	147	
2.34	66620	22420	17.5	10.1	3331	1121	3672	1700	0.850	18.2	8.05	303	172	
2.35	78040	26200	17.5	10.1	3844	1300	4280	1977	0.846	16.1	9.56	462	200	
2.37	92770	31030	17.7	10.2	4482	1532	5026	2331	0.848	14.1	11.6	714	232	
2.41	119200	39360	18.2	10.4	5570	1934	6311	2941	0.854	11.6	15.2	1317	283	
2.49	187100	60530	18.8	10.7	8172	2903	9540	4436	0.853	8.5	25.2	3885	415	
2.60	297900	94360	19.7	11.1	11960	4369	14460	6724	0.852	6.4	43.2	11060	605	